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THE DETERMINANTS OF TERRORISM IN WEST AFRICA: A SUBNATIONAL ACTOR-BASED ANALYSIS.

Bachelor Thesis

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Contents

	Abstract	6
	Introduction	8
1	Theoretical Framework and Methodology	10
1.1	Definition of Terrorism	10
1.2	Terrorism in West Africa	10
1.3	Problem Statement	13
1.4	Objectives of Study	14
1.5	Hypothesis	14
1.6	Literature Review	14
1.7	Research Methodology	16
2	Results	23
2.1	Baseline results	23
2.2	Results for Binomial logistic model with religious covariates	26
3	Conclusion	30

List of Figures

1.1	Yearly incidence from 2003 to 2022.	12
1.2	Yearly incidence by Islamic state from 2010 to 2022.	13
3.1	Study area	36
3.2	Gridded study area.	37
3.3	Spatial distribution of top 7 terrorist groups in West Africa.	37
3.4	Incidence of terrorist attacks by country from 2003 to 2022.	38
3.5	Incidence type by country.	38
3.6	Top 7 Terrorist organizations in West Africa.	39
3.7	Yearly Evolution of terror fatality.	39
3.8	Yearly Evolution of Boko haram.	40
3.9	Yearly Evolution of Communal Militia.	40
3.10	Yearly Evolution of JNIM.	41

List of Tables

1.1	Distribution of Event types and top 7 Terrorist groups.	11
1.2	Descriptive Statistics	19
2.1	Binomial logistic model estimation results.	24
2.2	Binomial logistic model estimation results.	25
2.3	Binomial logistic model with religious covariates estimation results (Odds Ratio).	26
2.4	Significance and relevance of the model	28
2.5	Model comparison by AIC	29
3.1	Details on variables used in regression	34
3.2	Confusion matrix for Ambazonian Separatists Source: Author	34
3.3	Confusion matrix for Boko Haram Source: Author	35
3.4	Confusion matrix for Communal Militia Source: Author	35
3.5	Confusion matrix for Fulani Militia Source: Author	35
3.6	Confusion matrix for Islamic State Source: Author	35
3.7	Confusion matrix for JNIM: Group for Support of Islam and Muslims Source: Author	35
3.8	Confusion matrix for Katiba Macina Source: Author	36

Acronyms

ACLED Armed Conflict Location and Event Data Project. 6, 7, 9, 10, 16, 20

AIC Akaike Information Criterion. 4, 9, 29

AQI Al-Qaeda in Iraq. 13

AUC Area Under the ROC Curve. 28

GDP Gross Domestic Product. 6, 9, 17, 19, 23–27, 30, 34

ISGS Islamic State in the Greater Sahara. 15

ISWAP Islamic State in West Africa Province. 8, 11

JNIM Jama’at Nasr al-Islam wal Muslimin. 9, 11, 15, 23, 26, 28, 29

OR Odds Ratio. 25, 27

PRIO The Peace Research Institute Oslo. 16

USA United States of America. 14

WA West Africa. 3, 6, 11–14, 23, 30, 37

Abstract

Following recent advancements in terrorism literature and the inclusion of spatial analysis, this paper seeks to study terrorism in West Africa from an actor-based perspective to understand the factors influencing terrorism in the region. We examine the heterogeneity of these factors for the top seven terrorist organizations operating actively in West Africa. We use a spatial domain represented by regular cells at 0.5° spatial resolution (55.5km x 55.5km). We approximate and deploy a binomial logistic model for each terrorist organization using data from the Armed Conflict Location and Event Data Project. This methodology allows for the close examination of each group, as well as all groups combined to understand how economic, political, demographic, and religious characteristics of locations influence the decision-making process of terrorist groups when deciding which areas to target. Our study demonstrates that GDP per capita, distance to the nearest national border, infant mortality rates, the share of Christians and Muslims, relative wealth index, and local government index remain the core factors influencing terrorism in the region. We find that the likelihood of attacks on cells reduces as GDP per capita rises. Additionally, we find that the probability of attacks also reduces as states develop higher levels of local government control. This finding is relevant for domestic policy within individual African countries as governments could attempt to build local government capacity, while simultaneously engaging in economic development strategies.

Keywords: Terrorism, West Africa, Spatial Analysis, Heterogeneity, Binomial Logistic Regression.

Résumé

Suite aux récents progrès de la littérature sur le terrorisme et à l'inclusion de l'analyse spatiale, cet article étudie le terrorisme en Afrique de l'Ouest du point de vue des acteurs pour comprendre les facteurs qui influencent le terrorisme dans la région. Nous examinons l'hétérogénéité de ces facteurs pour les sept principales organisations terroristes opérant activement en Afrique de l'Ouest. Nous utilisons un domaine spatial représenté par des cellules régulières à une résolution spatiale de $0,5^\circ$ (55,5 km x 55,5 km). Nous approximations le modèle logistique binomial pour chaque organisation terroriste utilisant les données d'Armed Conflict Location and Event Data Project (ACLED). Cette méthodologie permet d'examiner de près chaque groupe, ainsi que tous les groupes réunis pour comprendre comment les facteurs économiques, politiques, démographiques, et les caractéristiques religieuses des lieux influencent le processus de prise de décision des terroristes groupes au moment de décider quels domaines cibler. Notre étude démontre que le PIB par habitant, distance à la frontière nationale la plus proche, taux de mortalité infantile, proportion de chrétiens et de musulmans, l'indice de richesse relative et l'indice des collectivités locales restent les principaux facteurs influençant le terrorisme dans la région. Nous constatons que la probabilité d'attaques contre les cellules diminue à mesure que le PIB par habitant se lève. De plus, nous constatons que la probabilité d'attaques diminue également à mesure que les États se développent. niveaux plus élevés de contrôle du gouvernement local. Cette constatation est pertinente pour la politique intérieure au sein les pays africains individuels, car les gouvernements pourraient tenter de renforcer les capacités des gouvernements locaux, tout en s'engageant simultanément dans des stratégies de développement économique.

Mots clés: Terrorisme, Afrique de l'Ouest, analyse spatiale, régression logistique binomiale.

Introduction

The watch on terrorism in Africa continues as several countries, particularly in the Sahel region and the Horn of Africa, remain constantly challenged by terrorism. A comprehensive review of terrorism in West Africa reveals a refined narrative shaped by historical tensions and contemporary geopolitical influences. The turn of the 21st century witnessed a paradigm shift with the infiltration of jihadist ideologies, notably embodied by Al-Qaeda in the Islamic Maghreb (AQIM) operating in the Sahel ([Council on Foreign Relations, 2021](#)). The last decade marked the ascent of Boko Haram in Nigeria, a militant group with far-reaching consequences for regional security ([International Crisis Group, 2015](#)). Recent years have also seen the emergence of the Islamic State in West Africa Province (ISWAP), a splinter group from Boko Haram, further complicating the security landscape ([Institute for Security Studies, 2019](#)). Notably, the Global Terrorism Index for 2023 identifies four African countries among the top 10 nations most affected by terrorism ([Institute For Economics & Peace, 2023](#)). Burkina Faso ranked second, while Mali was at number 4. Nigeria and Niger are also among the top 10 impacted nations at number 8 and 10, respectively.

Studying terrorism in West Africa is of paramount importance given the specific and evolving challenges faced by the region. The geopolitical significance, coupled with the profound socio-economic and political impacts of terrorism, demands a focused and comprehensive understanding of the trends. Terrorism in West Africa has far-reaching socio-economic consequences. Attacks often result in displacement, disruption of essential services, and the loss of livelihoods for affected communities. The role of state institutions in addressing terrorism cannot be overstated. A comprehensive approach involves strengthening governance, promoting the rule of law, and engaging in community-based initiatives ([Jackson et al., 2011](#)). By studying terrorism in West Africa, policy-makers can better identify the weaknesses within state institutions and implement targeted reforms to enhance their effectiveness in preventing and responding to security threats. West Africa's complex landscape of terrorism involves the intersection of local insurgencies with global jihadist networks. The study of these connections is essential for recognizing and disrupting the transnational nature of extremist activities. It enables a more nu-

anced understanding of how international dynamics contribute to regional security challenges, emphasizing the need for coordinated efforts at both local and global levels (Le Sage, 2007).

The application and advancements of spatial analysis in terrorism literature have yielded valuable insights, incorporating the geographical attributes of locations into studies (Python, Andreas, et al., 2021; Python, Janine, et al., 2017; Walther et al., 2020). However, it focuses on aggregated terror incidences with no distinction to different terrorist organizations involved and ineluctably ignores the heterogeneity in the determinants of terrorism. One exception is Metiki (2021), that groups terrorist activities based on the ideology of the groups involved. However, his analysis does not include a spatial domain. Consequentially, there is a need to explore the diverse determinants of terrorist attacks specific to each organization.

In this study, we present a framework to explain the occurrence of terrorism perpetrated by the seven most prominent terrorist organizations in West Africa. Our analysis draws on data from the Armed Conflict Location and Event Data Project (ACLED) database for the period between 2003 and 2022, covering six West African countries. We include spatial data on economic, political, religious, demographic, and geographic variables, which may influence choice of target locations for terrorist groups. We extract all variables into a 0.5° spatial grid with R and create dummy variables for each terrorist organization and one dummy for all groups together. These variables are coded as "1" if that group attacked a cell and "0" if not. We fit the data on multiple independent binomial logistic regressions for each terrorist organization, which allows us to identify and compare the role of various terrorism drivers across different groups. The first model excludes variables of religious parity, while the second model includes them. The Akaike Information Criterion shows that all regression models with these religious variables explain more variation in the data. The findings highlight the significance of relative wealth, GDP per capita, population density, proximity to national borders and share of Christians and Muslims as factors influencing terrorist attacks in West Africa. Particularly, cells with higher GDP per capita have a lesser risk of terrorist attacks, confirming that poverty is a focus of terrorism in West African countries. An exception is the Jama'at Nasr al-Islam wal Muslimin, which is more likely to attack with a rise in cell GDP per capita. The odds that any group attacks reduce as local government control increases. Furthermore, the likelihood of attacks reduces for cells farther from the border. Except for JNIM, Communal militia and Fulani ethnic militias, which are more likely to attack cells farther from a national frontier.

The rest of the paper is structured as follows: Section 2 provides a descriptive analysis of terrorist activities by the top seven terrorist groups from 2003 to 2022 in West Africa and presents our data and empirical models. Section 3 presents the econometric results and summarizes them.

Theoretical Framework and Methodology

1.1 Definition of Terrorism

Terrorism encompasses acts of violence carried out by non-state organizations against representatives of a nation-state. It involves politically, religiously, or ethnically motivated attacks outside legitimate warfare perpetrated by non-state organizations targeting representatives of a nation-state to communicate to a large audience to achieve political, religious, or other goals. [Adu and Osadola \(2022\)](#) characterize terrorism as the unlawful use of force or violence against persons or property to threaten or coerce a government, the civilian population, or any part thereof, in order to achieve political or social objectives. Terrorism involves one violence-producing actor (for example, the Islamic State) and another actor who may or may not produce violence. For example, civilians attacked in an altercation cannot retaliate, but the military or police can attempt to contain the situation to the extent of their ability. It is important to note that these attacked actors must be representatives of a nation-state where the intended message passes through to. Specifically, representatives could range from civilians, military, and church buildings to state facilities. Data from ACLED is filtered to include only events that match the above definition of terrorism.

1.2 Terrorism in West Africa

Over the past few years, West Africa has witnessed a concerning evolution in the landscape of terrorism ([Blanchard, 2021](#)). In recent years, the region has experienced an upsurge in terrorist activities, with several groups gaining prominence. The multifaceted factors contributing to this phenomenon include economic grievances, political instability, and religious extremism ([United Nations Office for West Africa and the Sahel., 2020](#)).

In Burkina Faso, Islamist extremist groups, affiliated with both Al-Qaeda and the Islamic State, have intensified their attacks, targeting civilians, security forces, and governmental institutions. The northern regions, in particular, have become hotspots of violence, with an alarming increase in displacement and humanitarian crises

(International Crisis Group, 2020). Mali, situated in the Sahel region, has faced persistent violence from groups like Jama'at Nusrat al-Islam wal-Muslimin (JNIM). The central and northern regions, including areas such as Mopti and Timbuktu, have been focal points of insurgent activities. The complex ethnic and political landscape further complicates counterterrorism efforts, posing challenges to establishing lasting stability (BBC News, 2021). Benin, although not at the epicenter of terrorism, has faced challenges in maintaining security due to its proximity to high-risk areas especially in the northern part. In Nigeria, Boko Haram has continued to be a major player in the region, showcasing adaptability in tactics and strategies (Campbell, 2019). The northeastern regions of Nigeria, particularly Borno state, have borne the brunt of the insurgency. Additionally, the expansion of affiliated groups such as the Islamic State in West Africa Province (ISWAP) has further escalated the security crisis. The Sahel region, encompassing countries like Mali, Burkina Faso, and Niger, has become a hotspot for terrorist activities, posing a significant threat to regional stability.

Terrorist incidents in WA date back to the early 2000's, with a gradual rise starting in 2010 and an increment in the frequency and severity of attacks from 2014 onward. Table 1.1 shows that incidences from 2003 - 2007 made up 0.14%, from 2008 - 2012 made up 3.21%, from 2013 - 2017 made up 12.63%, and from 2018 - 2022 made up 84.03% of total incidence which stood at 15916. Violence against civilians remains a top activity for terrorists, especially in Nigeria, where it makes up 69% of all incidences recorded in the region. Notably, 51% of all incidences recorded at least one fatality. Table 1.1 depicts the top 7 terrorist organizations present in WA. The JNIM, an Islamist militant organization with missions to establish a strict form of Islamic law and governance in the areas it operates, represent (23.84%) of all incidences, and is present in Burkina Faso, Chad, Mali, and parts of Benin.

Table 1.1. Distribution of Event types and top 7 Terrorist groups.

	All events (%)	Benin (%)	Burkina Faso (%)	Cameroon (%)	Chad %	Mali (%)	Niger (%)	Nigeria (%)	Togo (%)
Event type									
Battles	20.4	46.15	15.51	28.86	19.47	13.88	41.18	19.4	37.5
Explosions/Remote violence	8.92	10.26	7.36	7.18	15.52	7.21	20.92	8.53	25
Strategic developments	14.52	15.38	32.11	15.81	15.31	22.08	2.61	2.81	.
Violence against civilians	56.16	28.21	45.02	48.14	49.7	56.83	35.29	69.26	37.5
Damage									
No fatalities	48.66	51.28	69.96	53.36	52.25	59.64	31.37	31.24	31.25
At least one	51.34	48.72	30.04	46.64	46.75	40.36	68.63	68.76	68.75
Yearly interval									
2003-2007	0.14	.	.	.	.	.	.	0.37	.
2008-2012	3.21	.	.	.	.	.	.	8.63	.
2013-2017	12.63	.	0.23	18.05	4.47	5.99	20.55	20.89	.
2018-2022	84.03	100	99.77	81.95	95.53	94.01	79.45	70.11	100
Top terror groups									
Ambazonian Separatists	7.83	.	.	99.84	.	.	.	0.16	.
Boko Haram	8.02	.	.	13.56	.	11.29	0.78	74.37	.
Communal Militia	11.77	.	.	.	.	.	.	100	.
Fulani Militia	10.57	.	.	0.06	9.63	.	.	90.31	.
Islamic State	34.75	0.11	10.5	33.25	6.91	17.25	2.59	29.4	.

Continued on next page

Table 1.1 – Continued from previous page

	All events (%)	Benin (%)	Burkina Faso (%)	Cameroon (%)	Chad %	Mali (%)	Niger (%)	Nigeria (%)	Togo (%)
JNIM	23.84	0.84	63.82	.	29.12	5.8	.	.	0.42
Katiba Macina	3.22	0.19	36.84	.	62.96	.	.	.	.
Total no incidents	15916	39	3192	3257	153	1972	1318	5969	16
Share in total incidents	100	0.25	20.05	20.46	0.96	12.39	8.28	37.50	0.10

Source: Author

The Islamic State, a jihadist extremist group with a mission to establish an Islamic caliphate in WA, represents (34.75%) of all incidences, and is operative in Nigeria, Niger, Chad, and Cameroon. Boko Haram, established to enforce a strict form of Islamic law in the region and to create an Islamic state governed by their interpretation of Sharia law, represents (8.02%), and is present in Cameroon, Mali, and Niger while dominating in Nigeria. The Fulani ethnic militia, present in Nigeria and Chad, make up 10.57% of all incidences recorded. Leading in insurgences, Nigeria records the most terrorist incidences in the region at 37.5% of all recorded occurrences. Cameroon, Burkina Faso and Mali follow behind with 20.46%, 20.05%, and 12.39%, respectively, of all recorded occurrences.

Figure 1.1 shows the yearly evolution of terrorist activities in WA from 2003 to 2022. It depicts that early occurrences of terrorism started in the North-Eastern part of Nigeria in the late 2000's and began spreading across the region in 2014 and worsens overtime. The figure shows the spread of terrorist activities from Nigeria to the borders of Cameroon and Chad, and later from 2014 onwards, to Mali, Burkina Faso and Niger.



Figure 1.1. Yearly incidence from 2003 to 2022.

Source: Author

The division between “domestic” and “international” terrorism corresponds to cases where the members of the organization attack representatives of a state with which they share a territorial or political affiliation

or attack representatives of foreign states (Enders and Sandler, 2002). An example is the Islamic State which originated in Iraq and was initially called Al-Qaeda in Iraq (AQI) and now has active subdivisions in Cameroon, Chad, Nigeria, Niger, Burkina Faso, and Mali.



Figure 1.2. Yearly incidence by Islamic state from 2010 to 2022.

Source: Author

Figure 1.2 shows that early activities of the Islamic State started in northern Nigeria in 2015, but has gradually spread across the region to Mali, Niger, Cameroon, Chad and Burkina Faso. This stands as evidence that terrorist organizations in WA enjoy operational success across the region.

1.3 Problem Statement

In recent years, West Africa has experienced a distressing surge in terrorism, resulting in a significant loss of life and property. The Sahel region in sub-Saharan Africa is now the epicentre of terrorism, with the Sahel accounting for more terrorism deaths in 2022 than both South Asia and the Middle East and North Africa (MENA) combined. Deaths in the Sahel constituted 43 per cent of the global total in 2022, compared to just one per cent in 2007 [Institute For Economics & Peace \(2023\)](#). These figures underscore the urgent need for a comprehensive examination of the factors influencing the occurrence of terrorism in West Africa (WA).

Terrorism poses a threat to lives and properties, prevents economic growth and foreign investment, significantly impacts the tourism industry and causes long-term economic stagnation. The core problem addressed by this research is to unravel the heterogeneity of factors that contribute to the decision-making processes of

terrorist groups operating in West Africa. By closely examining economic, political, demographic, and religious variables at the spatial level, this paper seeks to identify key determinants shaping the selection of target areas by these terrorist organizations.

Research Question

We aim to add to the existing literature by answering the following question: How do economic, political, demographic, and religious characteristics of locations influence the decision-making process of terrorist groups when deciding which areas to target?

1.4 Objectives of Study

The main objective of this study is to contribute to the existing literature on terrorism through a better understanding of the patterns of attacks from different terrorist groups. Specifically, this study aims to: determine the factors influencing the choice of target locations for the top seven terrorist organizations in West Africa.

1.5 Hypothesis

The following hypothesis has been formulated for our analysis:

- **The factors influencing terrorist attacks will differ in each group**

We consider this hypothesis because of the study of [Metiki \(2021\)](#) and [Kis-Katos, Liebert, and Günther \(2011\)](#). In fact, they argued that terrorism in sub-Saharan Africa is a heterogeneous phenomenon and found that determinants of terrorism differ according to the ideology advocated by the different terrorist groups. If the determinants of terrorism is heterogeneous across terrorist group ideologies, then these determinants are expected to be heterogeneous across individual terrorist groups in West Africa. As we seek to understand the patterns of attacks of the top 7 terrorist groups in West Africa, we expect that the factors influencing these attacks will differ in each terrorist group.

1.6 Literature Review

The watch on terrorism in Africa continues as several countries, particularly in the Sahel region and the Horn of Africa, remain constantly challenged by terrorism. Since the wake of the 7/11 attack on the United States of America, the international community has taken several actions against the rise of world terror. In the first 100 days after the drastic event that shook the hearts of many and took the lives of many, the United States began to destroy Al-Qaeda's grip on Afghanistan by driving the Taliban from power, disrupting its global operations and financing networks ([United States Department of States, 2001](#)). The USA began a worldwide hunt for terrorists with the help of allied nations. They launched a military campaign to suppress any ongoing terrorist operation. These ongoing complications led to various coalitions, changes in territory, and recruitment for many terrorist

organizations in more conducive areas like Africa. [Falode \(2016\)](#) posits that passage of leadership in terrorist organizations can often lead to changes in objectives, methods and general reasoning.

Terrorist activities have gone on for more than eight years in the Sahel, where Jama'at Nasr al-Islam wal Muslimin (JNIM) and the Islamic State in the Greater Sahara (ISGS) are driving sub-regional insurgencies in Mali, Burkina Faso, and Niger. After the emergence of ISGS in 2015, the two rival jihadi franchises for several years existed within an uneasy alliance in the region ([Héni and Caleb, 2020](#)). In 2020, the United Nations monitoring team that tracks the global jihadi threat warned the Security Council that the Islamic State franchises in West Africa and the Sahel continued to enjoy operational success in early 2020, as have those of Al-Qaida, heightening international concern about stability in the region ([Héni and Caleb, 2020](#)). [Hamidu \(2022\)](#) Suggests that terrorist organizations could also exploit border insecurity present in the region as opportunities to recruit, camp, radicalize youths both within and across, and execute their plans against their enemies/targets. [Adu and Osadola \(2022\)](#) argue that although identity politics plays a trivial role in the survival, peace, and continued existence of independent African states, the ideology of good neighbourliness fueled by identity politics has led to the spread of insurgence across international borders and heightened cooperation between groups within the region.

The factors affecting terrorism are multifaceted and may include economic drivers like national income levels. Particularly, [Metiki \(2021\)](#) finds that terrorist attacks increase as we move from countries with high-income quartiles to countries with low-income quartiles and confirms that poverty is a breathing ground for terrorism. [Daniel and Thomas \(2013\)](#) find that the causal relationship that exists between terrorism and economic growth is heterogeneous over time and across space. They argue that different causal mechanics across countries associated with country-specific factors, including the degree of politico-economic development, cultural affiliation, the intensity and persistence of terrorist activity, and political instability, govern its robustness to shocks from terrorism and the extent to which terrorism is motivated by economic factors.

The link between terrorism and economic growth is complex, and it varies with factors like development, culture, and regional instability, often driven by dissatisfaction and economic disparities, especially in Africa, where social conditions and ethnic or religious tensions contribute to extremism. [Graham, Brock, and Gregory \(2008\)](#) conclude that terrorism may be the chosen means of expressing dissatisfaction with the status quo. From an economic point of view, it would suggest that the incidence of terrorist events is associated with the degree of income inequality. [Midlarsky \(2011\)](#) discusses how societal conditions like inequalities, social grievances, and ethnic or religious tensions contribute to the emergence of political extremism and its escalation into mass violence. Terrorism motivated by political ideologies in Sub-Saharan African countries is low, whilst most groups in the region strive for the dominance of their ethnicity or the supremacy of their religion ([Metiki, 2021](#)).

Spatial analysis functions as a geospatial compass, guiding economists and data analysts through the intricate spatial domain of data. It lies at the intersection of geography, data science, and social sciences and plays a pivotal role in our understanding of complex spatial phenomena. Additionally, spatial analysis harnesses

the power of geospatial data and advanced computational techniques to unearth insights often hidden in the spatial patterns of our world. While the influence of geography on human agency is a subject of debate in scholarly discussions, the inclusion of location-based factors in these models acknowledges the inherent geographic variations in resources for political action. It also recognizes that some essential influences on human agency can be identified and quantified in the specific geographic contexts where violent events occur (Walther et al., 2020). Studies of social conflict using georeferenced conflict data have shown that cell phone coverage and environmental factors such as rainfall levels and weather conditions all increase the conflict propensity at the local level (Josiah et al., 2018). They argue that accessibility shapes the likelihood of a terror attack, and so targets near international borders or located in urban areas are more likely to be attacked. Both jihadist groups and criminal gangs prosper in areas lacking government control, often found in rural regions far from capital cities and close to national borders. The primary Islamist organizations, ISGS and JNIM, seek recruits and resources in these lawless areas, and they sometimes establish agreed-upon locations of influence, even engaging in fierce competition for the same territory (Orosz, 2022).

Choosing the appropriate grid size is crucial for effectively analyzing political violence across a vast region. A smaller 10km by 10km grid might offer a more detailed view, but it would often result in empty cells or cells with only a single event, making the analysis less meaningful. On the other hand, a larger 100 by 100 km grid could aggregate distant events that are not necessarily related. In contrast, a 50 by 50 km grid strikes a balance, providing enough events in each cell for meaningful analysis while offering a localized view of political violence. This approach facilitates comparisons of conflict evolution across a broad region, enabling a more consistent representation of violence compared to varying administrative units that differ in size significantly, especially in sparsely populated areas like the Sahara (Walther et al., 2020). Python, Andreas, et al. (2021) use regular cells at 0.5° spatial resolution to explain the occurrence of terrorism worldwide a week ahead and at a fine spatial scale, Walther et al. (2020) use 50km by 50km grid cells to present the spatial conflict indicator that measures conflict intensity and the spatial distribution of conflict within a region. Tollefsen et al. (2015) organize and maintain the PRIO-GRID dataset, a standardized spatial grid structure with global coverage at a resolution of 0.5 x 0.5 decimal degrees. Recent literature on terrorism incorporating a spatial domain supports the usage of grid cells sized at 0.5 x 0.5 decimal degrees (55km by 55km).

1.7 Research Methodology

Terrorism database

We gathered the data used in this study from secondary sources. The data on terrorism comes from the Armed Conflict Location and Event Data Project (Raleigh et al., 2010). Detailed information on the names of terrorist groups involved in any attack and the type of attack is well specified under the columns "actor1" and "event-type", respectively. They also include fatality counts and the geo-locations information in the latitude and longitude columns. We filtered the data to match the definition of terrorism employed in this paper.

We extracted the groups responsible for attacks in the study area from 2003 to 2022, the associated fatalities, and geo-locations. It represents a total of 15916 events for six countries. We dropped Benin and Togo because they recorded few incidences of terrorism. We also dropped unidentified groups and all terrorist groups that didn't meet our threshold of 500 total attacks from 2003 to 2022, reducing the number of active terrorist groups to 7 and the number of incidences to 15861.

Covariates

Economic covariates

We consider the role of the following economic factors:

- Gross Domestic Product (GDP) per capita from Global 1 km × 1 km gridded revised gross domestic product (Jiandong and Gao, 2021). Metiki (2021) posits that poverty is a focus of terrorism. We expect that the probability of an attack from any group should reduce with higher GDP per capita.
- Electricity consumption measures economic activity and is computed from night light data (Jiandong and Gao, 2021). We expect that the probability of an attack from any group should increase for low cell electricity consumption.
- The Relative Wealth Index measures the relative standard of living within countries using de-identified connectivity data, satellite imagery and other nontraditional data sources. The data is provided for 93 low and middle-income countries at 2.4km resolution by Chi et al. (2022).

Political covariates

We include the following variables from the variety of democracy dataset. Each of these selected variables range from 0 to 1 with "1" meaning the highest possible level has been attained and "0", no level attained. The dataset is a country-year datasets, so we extracted the following variables from 2003-2022:

- **Electoral democracy index:** which represents the extent of which the ideal of electoral democracy in its fullest sense is achieved in a country. This index is formed by taking the average of, on the one hand, the weighted average of the indices measuring freedom of association, clean elections, freedom of expression, elected officials, and suffrage and, on the other, the five-way multiplicative interaction between those indices Coppedge et al. (2023). It ranges from low to high (0-1).
- **Liberal democracy index:** which represents the importance of protecting individual and minority rights against the tyranny of the state and the tyranny of the majority. The liberal model takes a "negative" view of political power insofar as it judges the quality of democracy by the limits placed on government. This is achieved by constitutionally protected civil liberties, strong rule of law, an independent judiciary, and effective checks and balances that, together, limit the exercise of executive power. To make this a measure of liberal democracy, the index also takes the level of electoral democracy into account Coppedge et al.

(2023). It ranges from low to high (0-1).

- **Deliberative democracy index:** focuses on the decision-making process in a polity. A deliberative process occurs when public reasoning focuses on the common good and motivates political decisions. According to this principle, democracy requires more than an aggregation of existing preferences. There should also be respectful dialogue at all levels—from preference formation to final decision—among informed and competent participants who are open to persuasion. To make it a measure of not only the deliberative principle but also of democracy, the index also takes the level of electoral democracy into account Coppedge et al. (2023). It ranges from low to high (0-1).
- **Egalitarian democracy index:** Egalitarian democracy is achieved when the rights and freedoms of individuals are protected equally across all social groups, and resources are distributed equally across all social groups Coppedge et al. (2023). It ranges from low to high (0-1).
- **Equality before the law and individual liberty index:** measures the extent to which public administration enforces legislation transparently and impartially, and to what extent do citizens enjoy access to justice, secure property rights, freedom from forced labour, freedom of movement, physical integrity rights, and freedom of religion. The index is formed by taking the point estimates from a Bayesian factor analysis model of the indicators for rigorous and impartial public administration, transparent laws with predictable enforcement, access to justice for men/women, property rights for men/women, freedom from torture, freedom from political killings, from forced labor for men/women, freedom of religion, freedom of foreign movement, and freedom of domestic movement for men/women Coppedge et al. (2023). It ranges from low to high (0-1).
- **Local government index** measures the extent at which elected local government officials operate without interference from un-elected bodies at the local level. This index is formed by averaging the following indices: equality before the law and individual liberties, judicial constraints on the executive, and legislative constraints on the executive Coppedge et al. (2023). It ranges from low to high (0-1). The lowest score would be reserved for a country that has no elected local governments. A medium score would be accorded a country that has elected local governments but where those governments are subordinate to unelected officials at the local level perhaps appointed by a higher-level body. A high score would be accorded to a country in which local governments are elected and able to operate without restrictions from unelected actors at the local level with the exception of judicial bodies. Naturally, local governments remain subordinate to the regional and national governments.

Religious covariates

- We consider the role of religious division, proxied by the percentage of Muslims and Christians in each cell. We use Carl and Hunziker (2018) spatially interpolated data on ethnicity. Python, Janine, et al. (2017)

suggests that for ethnic conflict, terrorists may deliberately target ethnically polarised societies to raise ethnic tensions, which may lead to ethnic conflict. We assume the same for religious cases.

Demographic covariates

- We include data on infant mortality from the [Center for International Earth Science Information Network \(2021\)](#) high-resolution data on infant mortality. We relate poor economic conditions to high levels of infant mortality, so we expect that the probability of an attack from any group should increase with higher levels of infant mortality rate.
- We include data on population density from Gridded Population of the World V4 ([Center for International Earth Science Information Network, 2018](#)). They provide data on a five-year basis from 2000 to 2020. We expect the probability of an attack from any group should be higher in densely populated areas.
- We include data on road network density from the GLOBIO GRIP dataset ([Meijer et al., 2018](#)). The ease of movement in various areas may facilitate terrorist activities.

Geographic covariates

We consider the role of geographical variables: altitude, and distance to the nearest national border. [Nemeth, Mauslein, and Stapley \(2014\)](#) suggest the distance to a national frontier and elevation might impact terrorist activity. We extract altitude data from [NOAA National Geophysical Data Center \(2009\)](#) and [NOAA Technical Memorandum: Amante and Eakins \(2009\)](#), which provide altitude values at high spatial resolution (1 arc-minute grid).

Table 1.2. *Descriptive Statistics*

Variables	Min.	1st Qu.	Median	Mean	3rd Qu.	Max.	Missing	Unit
GDP per capita	0.000588	0.008997	0.012	0.137	0.069	27.725	3	1000 USD
Electricity consmp.	0	0	1356	5799	4093	939686	3	kilowatthours
Road density	0	6.167	51.361	64.733	96.389	1480.97	522	km/km2
Infant mortality	0.00	37.12	61.47	55.25	72.36	134.52	501	per 1000 live births
Population density	0.028	0.361	7.851	48.214	40.375	7575.438		per square kilometer
Relative Wealth Index	-1.094	-0.528	-0.456	-0.431	-0.369	0.865	13601	
Electoral democracy index	0.140	0.28	0.49	0.44	0.566	0.687		
Liberal democracy index	0.031	0.12	0.32	0.27	0.40	0.50		
Deliberative democracy index	0.100	0.135	0.362	0.331	0.465	0.574		
Egalitarian democracy index	0.069	0.183	0.294	0.274	0.386	0.456		
Equality before the law and individual liberty index	0.166	0.501	0.672	0.614	0.764	0.848		
Local government index	0.039	0.367	0.702	0.625	0.879	0.968		

Continued on next page

Table 1.2 – *Continued from previous page*

Variables	Min.	1st Qu.	Median	Mean	3rd Qu.	Max.	Missing	Unit
Elevation	-236.5	289.4	363.3	422.7	491.8	1962.9		metres above sea level
Distance	0.053	35.445	100.541	124.243	187.949	455.101		Kilometer
Share of Christians	0	0.006	0.020	0.088	0.053	0.848		% of population
Share of Muslims	0	0.050	0.960	0.648	0.983	0.994		% of population

Source: Author

Data Preparation

We filtered the Armed Conflict Location and Event Data Project data to include only events where the primary actor(actor 1) was a terrorist group and the actor on the receiving end(actor 2) were civilians. We exclude events where two groups clash or the police attacking terrorist groups. Subsequently, we excluded all terrorist groups whose number of incidents did not sum up 500. For the spatial domain, we generated a grid of regular cells at 0.5° spatial resolution from the study area and obtained 1950 equally-sized square grid cells of 55.5km x 55.5km. We spatially joined the terrorism dataset and the grid together and then aggregated by year and grid ID. The resulting dataset was a long panel data containing, for each cell and each year, the terrorist groups that attacked, the number of times in that year which the group struck and the total fatality from that group within that year. We extracted all variables into the grid. Subsequently, we merged the two datasets by grid ID and year. The output is a long panel data frame containing the grid ID, year, terrorist group, count, fatality and 16 explanatory variables.

Outcome Variable

We have 8 outcome variables. The first outcome variable represents terror attack which is coded as "1" if the cell had any terrorist attack in that particular year and "0" if there was no attack. The 7 left represent each terrorist group and each is coded as "1" if a cell was attacked by that organization and "0" if there was no attack. We created dummy variables for each terrorist group and one variable representing all groups, where;

$$group_variable = \begin{cases} 1, & \text{if group attacked in cell} \\ 0, & \text{otherwise} \end{cases} \quad (1.1)$$

Model selection

The nature of the dependent variable is binary, with $y_i \in]0, 1[$. An appropriate approach is the binomial logistic model which is a statistical model used to analyze the relationship between one or more independent variables and a binary dependent variable. We wish to estimate the unknown probability that an individual cell i is attacked by any group given a set of independent variables $X_1, \dots, X_n$:

$$p(x) = \mathbb{P}(\{Y = 1\} | \{(X_1, \dots, X_n) = x\}) \quad \text{with} \quad x = (x_1, \dots, x_p), \quad (1.2)$$

The logit transformation function:

$$\text{logit}(y) = \ln \left(\frac{y}{1-y} \right) \in \mathbb{R}, \quad y \in]0, 1[\quad (1.3)$$

Its inverse is the function

$$\text{logit}^{-1}(y) = \frac{\exp(y)}{1 + \exp(y)} \in]0, 1[, \quad y \in \mathbb{R} \quad (1.4)$$

The binary logistic regression model is characterized by:

$\forall i \in 1, \dots, n$

- $(x_{1,i}, \dots, x_{p,i})$ is a realization of the random real vector $(X_1, \dots, X_p)$;
- Knowing $(X_1, \dots, X_n) = (x_{1,i}, \dots, x_{p,i})$, x_i, y_i is a realization of:

$$Y_i \sim \mathcal{B}(p(x_i)), \quad p(x_i) = \mathbb{P}(\{Y = 1\} | \{(X_1, \dots, X_p) = x_i\}), \quad (1.5)$$

- We have:

$$\text{logit}(p(x)) = \beta_0 + \beta_1 x_1 + \dots + \beta_p x_p, \quad x = (x_1, \dots, x_p), \quad (1.6)$$

where $\beta_0, \dots, \beta_p$ are unknown real coefficients.

Thus, $p(x)$ and $\beta_0 + \beta_1 x_1 + \dots + \beta_p x_p$ are linked by a logit transformation called the logit link.

We deduce the expression of $p(x)$:

$$p(x) = \text{logit}^{-1}(\beta_0 + \beta_1 x_1 + \dots + \beta_p x_p) = \frac{\exp(\beta_0 + \beta_1 x_1 + \dots + \beta_p x_p)}{1 + \exp(\beta_0 + \beta_1 x_1 + \dots + \beta_p x_p)} \quad (1.7)$$

We estimate $\beta_0, \dots, \beta_p$ from data by calculating $p(x)$ by substitution. To do that, the model uses the maximum likelihood method.

- **Likelihood**

Let $\beta = (\beta_0, \dots, \beta_p)$. The likelihood associated to $(Y_1, \dots, Y_n)$ is:

$$L(\beta, z) = \prod_{i=1}^n p(x_i)^{z_i} (1 - p(x_i))^{1-z_i}, \quad z = (z_1, \dots, z_n) \in \{0, 1\}^n \quad (1.8)$$

with $x_i = (x_{1,i}, \dots, x_{p,i})$.

- **Maximum likelihood estimation**

Maximum likelihood estimators of $\beta_0, \dots, \beta_p$, noted $\hat{\beta}_0, \dots, \hat{\beta}_p$, satisfy the following equations:

$$\sum_{i=1}^n (y_i - \text{logit}^{-1}(\hat{\beta}_0 + \hat{\beta}_1 x_{1,i} + \dots + \hat{\beta}_p x_{p,i})) = 0 \quad (1.9)$$

and $\forall j \in \{1, \dots, p\}$,

$$\sum_{i=1}^n x_{i,j} (y_i - \text{logit}^{-1}(\hat{\beta}_0 + \hat{\beta}_1 x_{1,i} + \dots + \hat{\beta}_p x_{p,i})) = 0 \quad (1.10)$$

2.1 Baseline results

In this section, we present the very first estimation results. Here, we exclude the variables representing religious division from the model. In Table 2.1 and Table 2.3, we have grouped the estimation results of the variables often used in the literature on terrorism. Each column shows the results for one of the seven identified terrorist organizations, with the first column for all groups combined. All models are estimated using a binomial logistic regression. They include sets of year dummies and region dummies. Incorporating year fixed effects and country fixed effects in our model is crucial, particularly because we are working with panel data that exhibits variation across time and among different countries.

Table 2.1 presents estimation results for the Binomial Logistic model. The analysis of Table 2.1 provides a first key piece of information: GDP per capita is significant at the 10% level. This means that GDP per capita is correlated with terror attacks in WA. The negative sign on the coefficient (-0.101) indicates that cells with higher GDP per capita have lesser risks of terrorist attacks, confirming that low national income can be associated to terrorist activities in West African countries. With an exception to JNIM (significant at the 5% level), whose chances of attack increases at places with higher GDP per capita. Population density is also significant at 10% level of significance for all identified, but at 1% for the Ambazonian Separatists and Katiba Macina. The likelihood of terrorist attacks reduces for densely populated cells. However, these chances increase for the Katiba Macina, Fulani, and Communal militia individually, who are more likely to attack when cell population density increases.

Table 2.1. Binomial logistic model estimation results.

Variables	All identified	Ambazonian Separatists	Boko Haram	Communal Militia	Fulani Ethnic Militia	Islamic State	JNIM	Katiba Macina
GDP per capita	−0.101*** (0.034)	0.454 (0.485)	−0.005 (0.143)	−0.018 (0.220)	−0.013 (0.103)	−0.056 (0.167)	15.906** (7.536)	−5.870 (5.974)
population density	−0.001*** (0.0002)	−0.009* (0.005)	−0.004*** (0.001)	0.003*** (0.001)	0.005*** (0.001)	−0.004*** (0.001)	0.0002 (0.003)	0.008* (0.004)
Democracy index	0.202 (0.451)	−24.559 (25.097)	−4.532*** (1.602)	13.013** (5.541)	9.062*** (1.535)	−2.677*** (0.818)	1.123 (1.004)	5.276** (2.385)
Local government index	−0.856*** (0.207)	2.530 (5.725)	0.864 (0.562)	1.142 (1.139)	0.543 (0.580)	−0.117 (0.359)	−3.115*** (0.992)	11.330*** (3.133)
Distance to border	−0.002*** (0.0003)	0.003 (0.007)	−0.00001*** (0.00000)	0.004*** (0.001)	0.005*** (0.001)	−0.005 (0.001)	0.009*** (0.002)	−0.008*** (0.003)
Elevation	0.0001 (0.0002)	0.002** (0.001)	0.001*** (0.0005)	0.002*** (0.001)	−0.002*** (0.0004)	−0.0003 (0.0004)	−0.007*** (0.002)	0.009*** (0.003)
Infant mortality	−0.002** (0.001)	−0.039*** (0.012)	0.001 (0.001)	0.004 (0.004)	−0.012*** (0.002)	0.017*** (0.002)	0.002 (0.003)	−0.011** (0.004)
Road density	−0.0005 (0.0003)	0.003 (0.003)	0.005 (0.001)	−0.003 (0.002)	−0.010*** (0.001)	0.003*** (0.001)	0.002 (0.002)	−0.003 (0.003)
Rel. Wealth Index	2.428*** (0.232)	3.571 (3.151)	3.519*** (0.833)	−7.995*** (1.066)	−0.381 (0.635)	0.702 (0.565)	−0.018 (0.981)	1.503 (1.533)
AUC	0.5	0.94	0.75	0.69	0.76	0.69	0.85	0.59
No. observations	25,233	2,487	2,487	2,487	2,487	2,487	2,487	2,487
No. attacks	15916	1246	1276	1873	1682	5531	3795	513
Year fixed-effect	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Country fixed-effect	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

All models are estimated using a binomial logistic regression. They include sets of year dummies and region dummies.

Estimation results are presented in log odds, with standard errors in parentheses.

For the sake of readability, statistically significant coefficients are with bold stars

***Significant at the 1% level

**Significant at the 5% level

*Significant at the 10% level

Source: Author

Democracy's impact on the likelihood of attack varies among group. The democracy index is insignificant for all aggregate groups but gains significance for groups individually. The index is significant at the 5% level for Katiba Macina and Communal militia but significant at the 10% level for Boko Haram, Fulani Militia and Islamic State. The chances of Boko Haram and Islamic State attacking reduce for higher levels of democracy. Also, the

Communal militia, Fulani ethnic militia, and Katiba Macina increase the chances of attacks for cells with a higher democracy index.

Table 2.2 shows the Odds Ratio for Table 2.1 for better interpretations of the model results.

Table 2.2. *Binomial logistic model estimation results.*

Variables	All identified	Ambazonian Separatists	Boko Haram	Communal Militia	Fulani Ethnic Militia	Islamic State	JNIM	Katiba Macina
GDP per capita	0.904*** (0.034)	1.575 (0.485)	0.995 (0.143)	0.983 (0.220)	0.987 (0.103)	0.945 (0.167)	8,089,988.000** (7.536)	0.003 (5.974)
population density	0.999*** (0.0002)	0.991* (0.005)	0.996*** (0.001)	1.003*** (0.001)	1.005*** (0.001)	0.996*** (0.001)	1.000 (0.003)	1.008* (0.004)
Democracy index	1.223 (0.451)	0.000 (25.097)	0.011*** (1.602)	448,388.700** (5.541)	8,619.729*** (1.535)	0.069*** (0.818)	3.073 (1.004)	195.551** (2.385)
Local government index	0.425*** (0.207)	12.557 (5.725)	2.101 (0.562)	3.132 (1.139)	1.721 (0.580)	0.890 (0.359)	0.044*** (0.992)	83,293.210*** (3.133)
Distance to border	0.998*** (0.0003)	1.003 (0.007)	0.988*** (0.00000)	1.004*** (0.001)	1.005*** (0.001)	0.995*** (0.001)	1.009*** (0.002)	0.992*** (0.003)
Elevation	1.000 (0.0002)	1.002** (0.001)	1.001** (0.0005)	1.002*** (0.001)	0.998*** (0.0004)	1.000 (0.0004)	0.993*** (0.002)	1.009*** (0.003)
Infant mortality	0.998** (0.001)	0.962*** (0.012)	1.011** (0.001)	1.004 (0.004)	0.988*** (0.002)	1.017*** (0.002)	1.002 (0.003)	0.989** (0.004)
Road density	1.000 (0.0003)	1.003 (0.003)	1.005*** (0.001)	0.997 (0.002)	0.990*** (0.001)	1.003*** (0.001)	1.002 (0.002)	0.997 (0.003)
Rel. Wealth Index	11.337*** (0.232)	35.541 (3.151)	39.811*** (0.833)	0.0003*** (1.066)	0.683 (0.635)	2.018 (0.565)	0.982 (0.981)	4.497 (1.533)
AUC	0.5	0.94	0.75	0.69	0.76	0.69	0.85	0.59
No. observations	25,233	2,487	2,487	2,487	2,487	2,487	2,487	2,487
No. attacks	15916	1246	1276	1873	1682	5531	3795	513
Year fixed-effect	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Country fixed-effect	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

All models are estimated using a binomial logistic regression. They include sets of year dummies and region dummies.

Estimation results are presented in Odds Ratio, with standard errors in parentheses.

For the sake of readability, statistically significant coefficients are with bold stars

***Significant at the 1% level

**Significant at the 5% level

*Significant at the 10% level

Source: Author

Proximity to national borders increases the likelihood of terrorist attacks, with exceptions for certain groups. From Table 2.2, we confirm that areas closer to a national border have higher chances of becoming targets. Notably, the odds of a group targeting cells decreases by 0.2% for an additional kilometer away from a national frontier. However, the opposite occurs for communal militias, Fulani ethnic militia and the JNIM, who are more likely to attack cells farther from the border. Results also show that the likelihood of attacks reduces as local government control increases. The relative wealth index measures the standard of living of cells compared to other surrounding cells. The odds that a cell is attacked increases by 11.337 times for each additional unit of the relative wealth. Analysis shows that discontent and dissatisfaction with economic conditions could also increase the likelihood of terrorist attacks.

Boko Haram is less likely to attack densely populated cells and cells with high democracy index. Areas farther from the national border are less likely to be targeted as well. Results show that the likelihood of attacks from Boko Haram increases for cells with high infant mortality. Confirming that they are more likely to attack areas with poor economic conditions. The Islamic State reduces odds of targeting cells by 93%, with every additional unit of the democracy index. Suggesting that the Islamic State targets areas where democracy isn't popular. They also target areas closer to a national border as the odds reduce by 0.5% for an additional kilometre farther from the national frontier. The JNIM are less likely to attack areas with high local government control. Also, they are more likely to attack areas farther from a national frontier.

2.2 Results for Binomial logistic model with religious covariates

Table 2.3 shows the results for the binomial logistic model with two additional variables added to the regression. These variables represent cell share of Christians and Muslims and may help us understand the religious dynamics of terrorism in the region.

Table 2.3. *Binomial logistic model with religious covariates estimation results (Odds Ratio).*

Variables	All identified	Ambazonian Separatists	Boko Haram	Communal Militia	Fulani Ethnic Militia	Islamic State	JNIM	Katiba Macina
GDP per capita	1.003 (0.033)	0.910 (2.112)	1.139 (0.170)	1.163 (0.317)	0.831 (0.131)	1.335* (0.160)	248,882.700 (8.803)	0.00002 (6.865)
population density	0.999*** (0.0002)	0.987* (0.007)	0.998** (0.001)	1.005*** (0.001)	1.003*** (0.001)	0.997*** (0.001)	1.000 (0.003)	1.007* (0.004)
Democracy index	0.813 (0.449)	0.000 (49.714)	0.027** (1.600)	601,412.100** (5.453)	761.732*** (1.635)	0.156** (0.900)	1.542 (1.064)	12.206 (2.446)
Local government index	0.613** (0.209)	307,887.100 (10.062)	1.583 (0.580)	3.509 (1.134)	2.459 (0.671)	0.496* (0.396)	0.069*** (1.014)	56,561.410*** (3.026)

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Table 2.3 – Continued from previous page

Variables	All identified	Ambazonian Separatists	Boko Haram	Communal Militia	Fulani Ethnic Militia	Islamic State	JNIM	Katiba Macina
Distance to border	0.998*** (0.0004)	0.988 (0.013)	0.991*** (0.002)	1.007*** (0.001)	1.002** (0.001)	0.997*** (0.001)	1.010*** (0.002)	0.992*** (0.003)
Elevation	1.000* (0.0002)	1.005** (0.002)	1.001 (0.001)	1.002*** (0.001)	0.999** (0.0004)	1.000 (0.001)	0.993*** (0.002)	1.009*** (0.003)
Infant mortality	0.998* (0.001)	0.991 (0.021)	1.003 (0.005)	.001 (0.004)	0.992*** (0.003)	1.011*** (0.002)	1.003 (0.003)	0.992* (0.004)
Road density	0.999** (0.0003)	1.012 (0.007)	1.002 (0.001)	0.992*** (0.002)	0.997** (0.001)	1.000 (0.001)	1.002 (0.002)	0.998 (0.003)
Rel. Wealth Index	17.381*** (0.240)	876.349 (5.531)	209.982*** (0.908)	0.004*** (1.072)	0.037*** (0.806)	4.571** (0.600)	1.255 (0.983)	10.730 (1.583)
Share of Christians	10.182*** (0.228)	772.456* (4.010)	0.269 (1.020)	0.703 (1.416)	4.937** (0.730)	0.014*** (0.706)	3.742* (0.752)	2.978*** (0.959)
Share of Muslims	6.767*** (0.158)	0.0003*** (1.909)	47.360*** (0.704)	137.789*** (0.818)	0.003*** (0.541)	42.799*** (0.591)	0.679 (0.979)	0.235 (1.056)
AUC	0.59	0.98	0.77	0.76	0.81	0.73	0.84	0.60
No. observations	25,233	2,487	2,487	2,487	2,487	2,487	2,487	2,487
No. attacks	15916	1246	1276	1873	1682	5531	3795	513
Year fixed-effect	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Country fixed-effect	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

All models are estimated using a binomial logistic regression. They include sets of year dummies and region dummies.

Estimation results are presented in Odds Ratio, with standard errors in parentheses.

For the sake of readability, statistically significant coefficients are with bold stars

***Significant at the 1% level

**Significant at the 5% level

*Significant at the 10% level

Source: Author

The religious composition of a cell significantly impacts the likelihood of terror attacks, with the odds increasing for cells with higher proportions of Christians and Muslims. The share of Christians and Muslims is significant to all identified groups together. Other variables are not highly affected after including these two variables to the model, although, GDP per capita is no longer significant in the model. Table 2.3 shows that the likelihood of attacks on cells increases as the share of Christians and Muslims increases. More specifically, the odds that a cell is attacked increases by 10 times, for a unit increase to the proportion of Christians in that cell. Also, the odds that a cell is attacked increases by 6 times, for an extra unit to the share of Muslims in that cell.

The Islamic State is likely to attack areas with a higher share of Muslims and less likely to attack more

Christian-occupied areas. The likelihood that the JNIM targets an area increases by 3 times for every additional unit to the share of Christians. Fulani militia increases the odds of targeting cells by 4 times with more units of the share of Christians in that cell and reduces by 99% for higher Muslim population. The odds that Boko Haram attacks a cell increases by 47 times for an additional unit to the share of Muslims in that cell. These results confirm that the distribution of religious affiliations in a cell affects the likelihood of terror attacks. Specifically, it highlights that groups like the Fulani militia, JNIM, Katiba Macina, and Ambazonian Separatists are more likely to target areas more populated by Christians. Additionally, Boko Haram, Communal Militia, and Islamic State are more likely to target Muslim-occupied areas.

Hypotheses checking

After careful examination of the factors affecting the occurrence of terrorism in West Africa, we accept the hypothesis proposed in section 2.2 of this paper. Specifically, we confirm that the determinants of terrorist attacks differ from group to group. For example, the odds that Boko Haram attacks a cell reduces by 0.9% for each kilometer added to cell distance from the nation border, while these odds increase by 1% for the JNIM. Also, the Fulani ethnic militia increases chances of attack for cells with higher shares of Christians, while the Islamic State reduces chances of attacks for cells with higher share of Christians.

Significance and relevance of the model

The Area Under the ROC Curve (AUC) is included on each table. The AUC is an aggregated metric that evaluates how well a logistic regression model classifies positive and negative outcomes. It is also used to evaluate the predictive performance of a model. The models for terror groups has an AUC ranging from 0.59 - 0.94, indicating good performances and improves after including variables on religious parity to 0.60-0.98.

Table 2.4. *Significance and relevance of the model*

Measure	Ambazonian Separatists	Boko Haram	Communal Militia	Fulani Ethnic Militia	Islamic State	JNIM	Katiba Macina
Model 1							
lr test	0.000 ***	0.000 ***	0.000 ***	0.000 ***	0.000 ***	0.000 ***	0.000 ***
HL test(p-value)	0.959	0.133	0.734	0.893	0.005	0.000	0.702
Error rate	0.008	0.069	0.073	0.135	0.187	0.131	0.056
Model 2							
lr test	0.000 ***	0.000 ***	0.000 ***	0.000 ***	0.000 ***	0.000 ***	0.000 ***
HL test(p-value)	0.991	0.1559	0.776	0.829	0.000	0.000	0.685
Error rate	0.008	0.067	0.066	0.094	0.168	0.133	0.054

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Table 2.4 – Continued from previous page

Variables	Ambazonian Separatists	Boko Haram	Communal Militia	Fulani Ethnic Militia	Islamic State	JNIM	Katiba Macina
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To evaluate the relevance of the logistic regression model, we use the Hosmer-Lemeshow test. The associated p-value uses the Chi-square law: if p-value > 0.05, we admit that the model is well suited to the data. The error rate is the proportion of predicted modalities which differ from the observed modalities. The closer it is to 0, the better the predictive quality of the model.

*** Significant at the 1% level

** Significant at the 5% level

* Significant at the 10% level

Source: Author

Table 2.5. Model comparison by AIC

Terrorist group	Model1	Model2
All identified	9962.6	9805
Ambazonian Separatists	212.97	111.74
Boko Haram	891.3	815.85
Communal Militia	777.62	708.53
Fulani Militia	1563.4	1245.3
Islamic State	2224.1	1979.4
JNIM	1295.8	1294.3
Katiba Macina	701.93	690.76

In the comparison of two models, the better model is the one with the smallest AIC value.

Source: Author

Conclusion

In conclusion, this study identified the multifaceted determinants of terrorism in West Africa, shedding light on the distinct risk factors associated with each terrorist organization analyzed in this research. We addressed the fundamental question of how economic, political, demographic, and religious variables influence the decision-making processes of terrorist groups when selecting their targets. Our examination has singled out seven prominent terrorist organizations operating within the region. Notably, we have discerned that the determinants of terrorism, encompassing GDP per capita, infant mortality rates, population density, democratic levels, local government control, relative wealth indices, and the proportions of Christians and Muslims, exhibit notable variations among these groups. As a result, we emphasize the imperative need to tailor counter-terrorism strategies to accommodate the unique characteristics, goals, and tactics of each terrorist organization. Such an approach is essential, given the dynamic and heterogeneous nature of terrorism in West Africa. Solutions aimed at reducing poverty, bridging the economic gap, and improving democracy and local government control might be effective against terrorism.

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Appendix

Table 3.1. *Details on variables used in regression*

Variable	Format	Year	Unit
GDP per capita	tif	2003-2019	cell
Population Density	asc	2000, 2005, 2010, 2015, 2020	cell
Infant Mortality Rate	tif	2015	cell
Relative Wealth Index	csv	2022	country
Varieties of Democracy	csv	2003 - 2022	country
Religious variable	asc	2003 - 2022	cell
Distance to closest national border	computed from data	2022	cell
Elevation	tif	2022	cell

Source: Author

Confusion Matrices

	0	1
0	2404	2
1	4	77

Table 3.2. *Confusion matrix for Ambazonian Separatists*
Source: Author

	0	1
0	2169	54
1	115	149

Table 3.3. *Confusion matrix for Boko Haram*
Source: Author

	0	1
0	2203	67
1	99	118

Table 3.4. *Confusion matrix for Communal Militia*
Source: Author

	0	1
0	1897	53
1	186	351

Table 3.5. *Confusion matrix for Fulani Militia*
Source: Author

	0	1
0	1718	121
1	347	351

Table 3.6. *Confusion matrix for Islamic State*
Source: Author

	0	1
0	1687	208
1	121	471

Table 3.7. *Confusion matrix for JNIM: Group for Support of Islam and Muslims*
Source: Author

	0	1
0	2320	19
1	112	36

Table 3.8. Confusion matrix for *Katiba Macina*
Source: Author

Countries and terrorist groups description

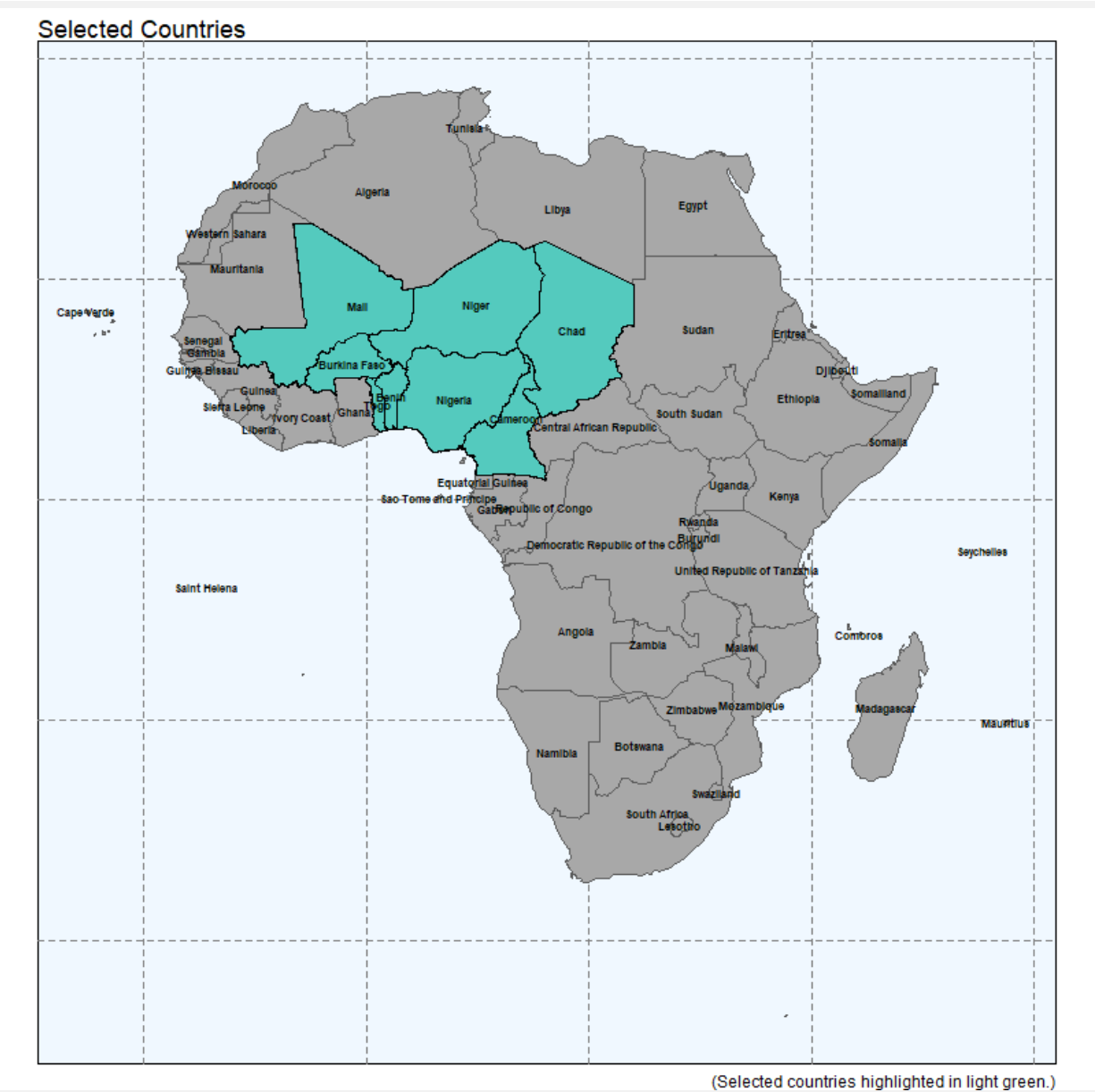


Figure 3.1. Study area

Source: Author

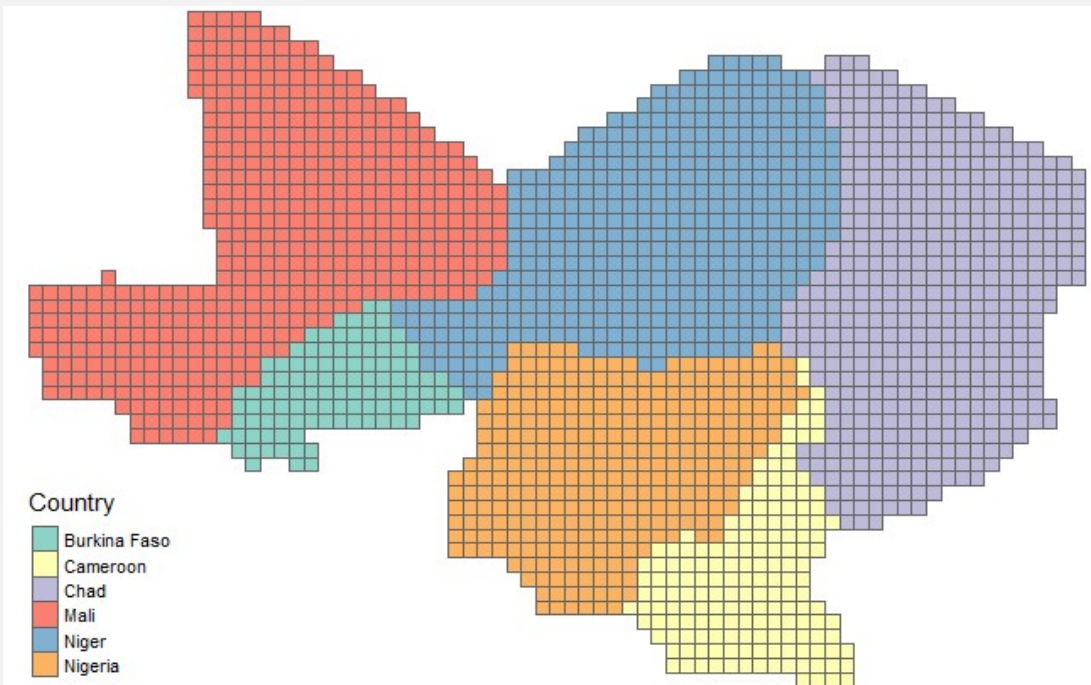


Figure 3.2. Gridded study area.

Source: Author

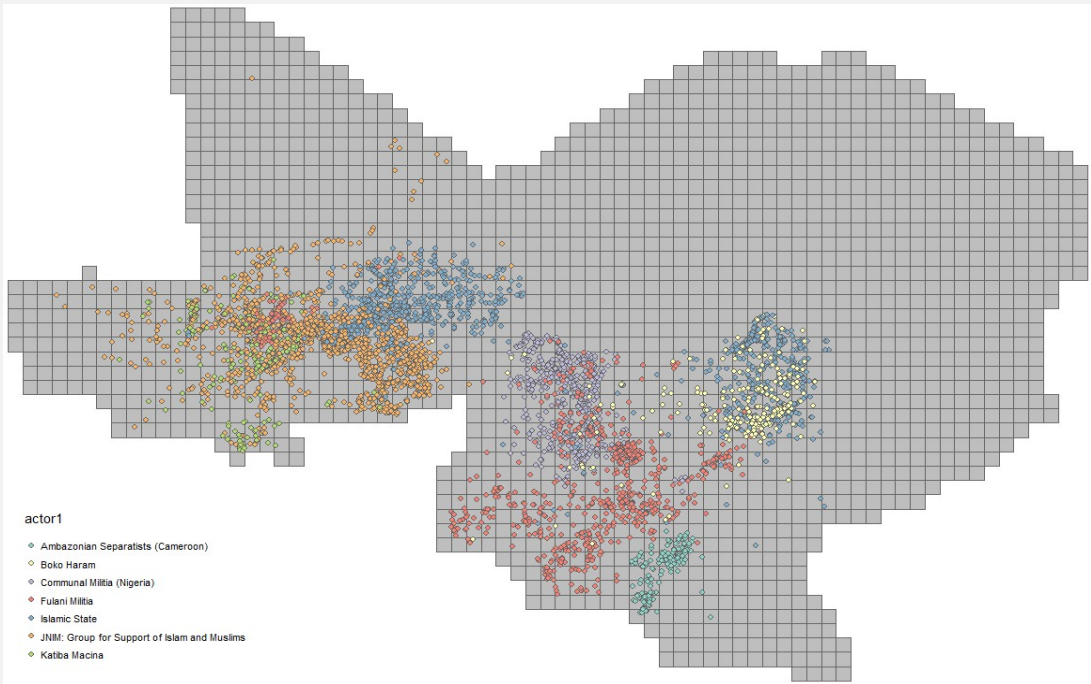


Figure 3.3. Spatial distribution of top 7 terrorist groups in West Africa.

Source: Author

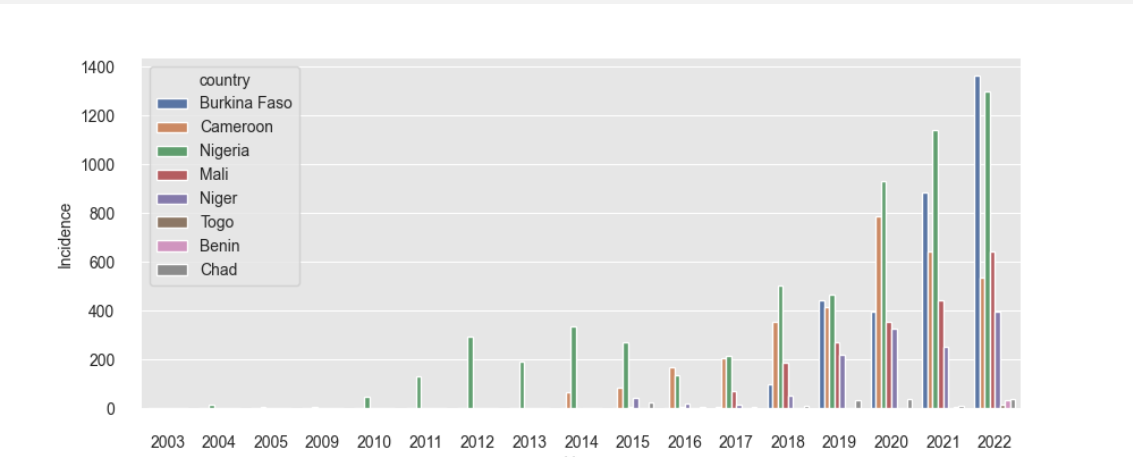


Figure 3.4. Incidence of terrorist attacks by country from 2003 to 2022.

Source: Author

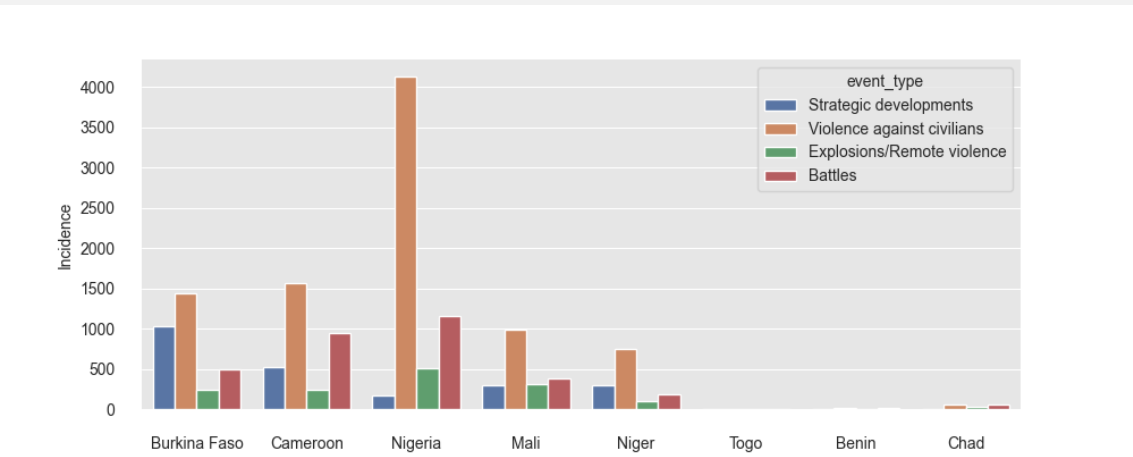


Figure 3.5. Incidence type by country.

Source: Author

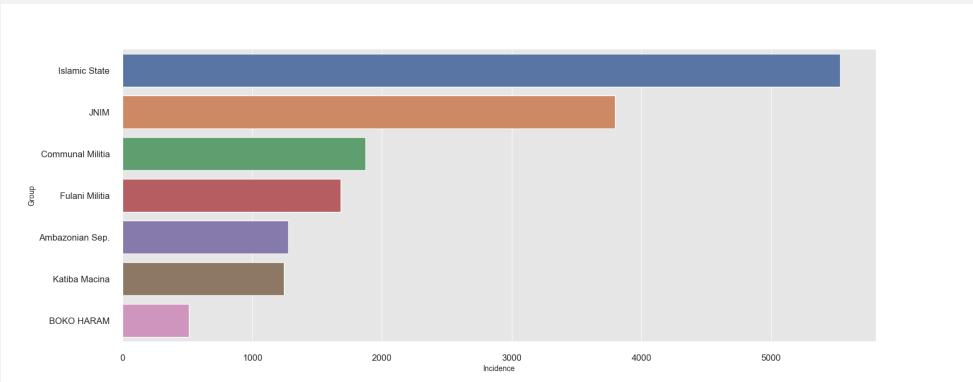


Figure 3.6. Top 7 Terrorist organizations in West Africa.

Source: Author



Figure 3.7. Yearly Evolution of terror fatality.

Source: Author

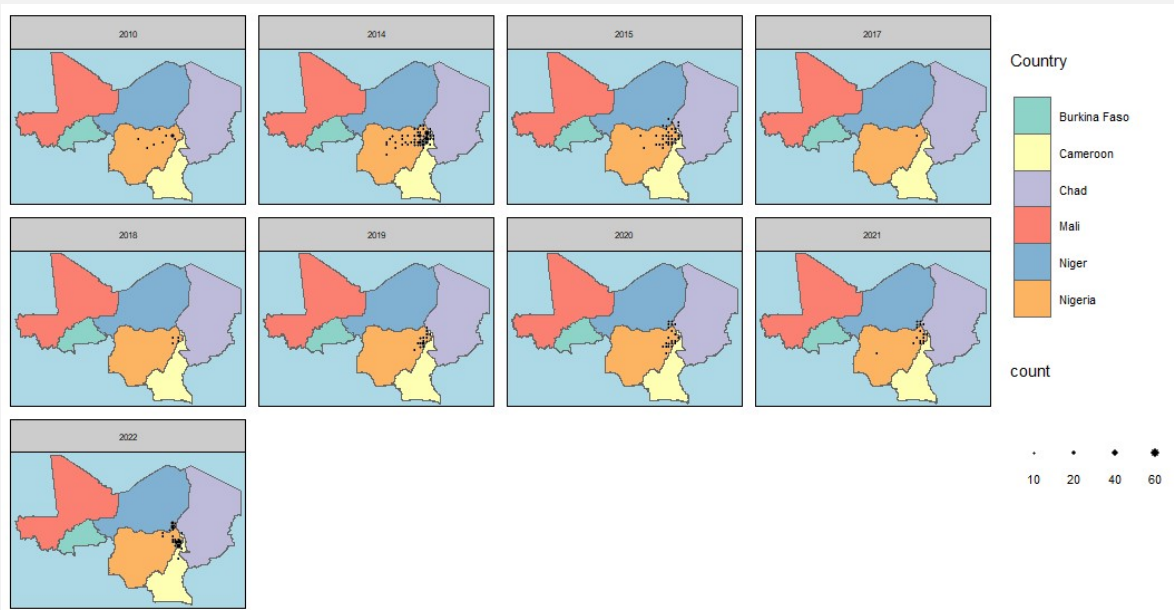


Figure 3.8. Yearly Evolution of Boko haram.

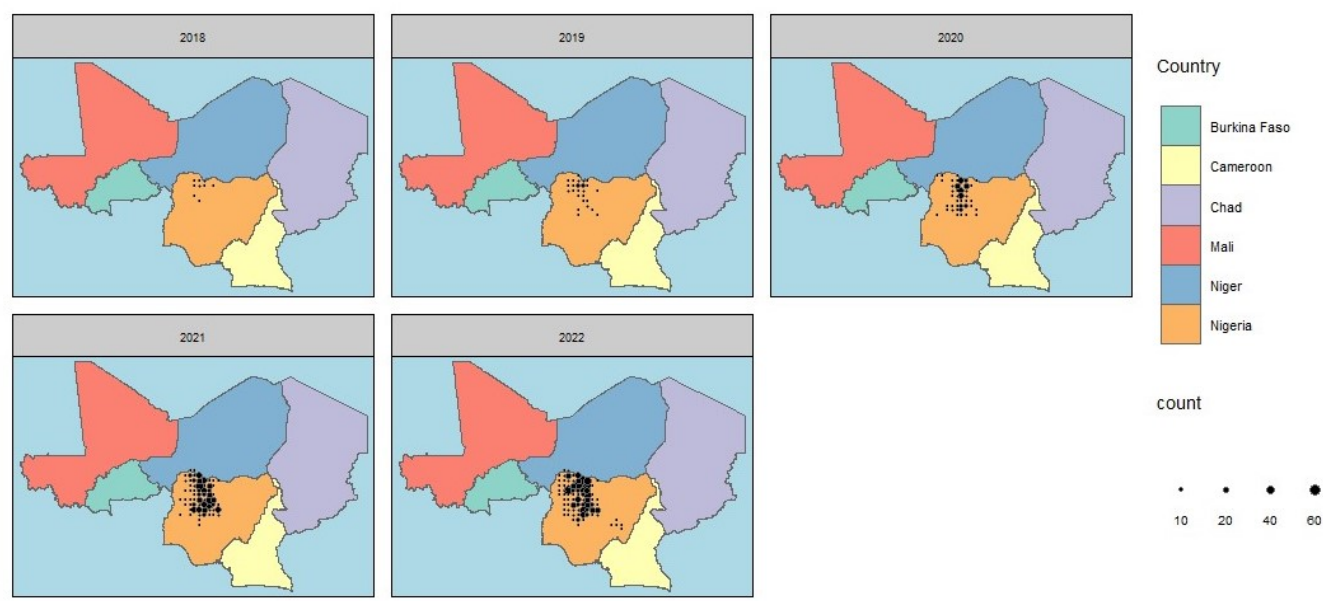
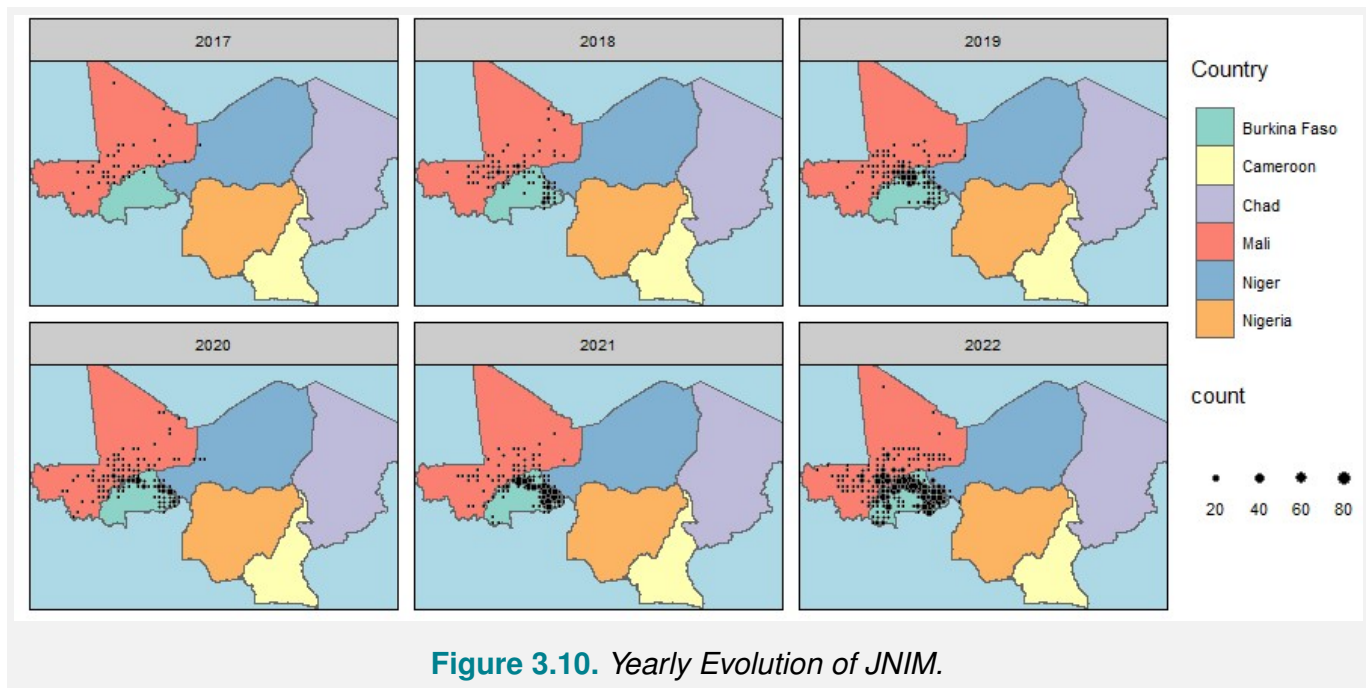


Figure 3.9. Yearly Evolution of Communal Militia.



Software used in producing this document includes; R, Python, Stata and $\text{\LaTeX}$. All materials used in the production of this thesis is available at <https://rb.gy/9ncfso>. Regression tables were generated using [Hlavac \(2022\)](#) stargazer package on R. Dummy variables were created using [Kaplan and Schlegel \(2023\)](#) fastDummies package on R.